PROJECT CASE STUDY

Pulse Robot

A responsive product-story website introducing Atlas, a humanoid companion concept, through immersive visuals, scroll-led storytelling, and interactive inquiry surfaces.

SYSTEM	SCOPE	STATUS
Product Marketing Website	**Frontend · Interaction · Responsive Product Story**	**Implemented Frontend Template**

01 The Problem

A humanoid-robot concept needs more than a list of capabilities: visitors need a clear introduction to what it represents, where it could fit, and how to explore it further.

Pulse Robot positions Atlas as an assistant for the spaces where people live and work. Its narrative connects repetitive tasks, human creativity, and safety with the promise of adaptive robotics. The web project translates that positioning into an approachable product story; it does not implement the robot itself.

> The website brings the introduction, product vision, displayed details, capability messaging, and inquiry interfaces into one continuous browsing experience.

02 What Was Built

A component-based frontend combining branded storytelling, responsive presentation, and browser-only interaction.

BRAND & PRODUCT INTRODUCTION

Introduce Pulse Robot and Atlas through a prominent hero, robot artwork, animated media, and a concise companion-focused proposition.

SCROLL-LED PRODUCT STORY

Move from a three-card vision sequence to displayed product details, an image showcase, capability cards, and testimonial copy.

RESPONSIVE EXPLORATION

Adapt the layout across screen sizes, switch to a mobile navigation menu, and provide smooth links to features and contact details.

INQUIRY & SIGNUP INTERFACES

Collect field input and demonstrate validation, loading, and toast feedback. These local demos do not send leads or subscriptions to a service.

03 How It Works

The browser renders a React application rather than contacting a product backend. Reusable sections compose the landing page; local state and browser events drive its interactive presentation.

Vite entry → React application → Routed landing page

Reusable sections · Local assets · Browser event handlers

Local state + scroll observers → Visual feedback and navigation

Entry & page composition

The HTML shell loads the React entry point. React Router renders the landing page at the root route and a not-found page for unmatched paths. The home page assembles the header, product sections, forms, and footer.

Media & scroll presentation

The hero loads a local Lottie animation and retains robot artwork as a fallback. Intersection observers trigger reveals; a scroll listener coordinated with requestAnimationFrame advances the sticky vision-card sequence.

Navigation & viewport behavior

The header responds to scroll position and manages a mobile menu with body-scroll locking. In-page links locate section IDs and scroll with a header offset. The hero also branches on viewport width for presentation effects.

Form interaction & feedback

The details form validates required input, shows a toast, and resets its fields. Newsletter signup uses a one-second simulated delay before feedback. Neither flow calls a submission API or persists the entered information.

04 Key Features

- **Responsive product landing page** — a single narrative connects the Atlas introduction, vision, details, artwork, capability messaging, and inquiry surfaces.
- **Desktop and mobile navigation** — a responsive header links to the product features and contact section, with a mobile menu and controlled page scrolling.
- **Animated hero presentation** — local Lottie media and a static-art fallback introduce the product, with desktop pointer and scroll effects in the hero code.
- **Scroll-driven vision sequence** — three stacked cards advance through a sticky section as visitors move through the product narrative.
- **Product-detail and capability presentation** — structured sections display specifications and six capability themes as promotional content, not verified robot functionality.
- **Demo and access inquiry interface** — name, email, and optional company fields provide local validation, success feedback, and a reset after submission.
- **Newsletter interaction** — an email field demonstrates a loading state and confirmation toast using a simulated request rather than a subscription service.
- **Reusable visual foundation** — section components, local imagery, Tailwind theme tokens, icons, and shared feedback primitives support a consistent presentation.

05 Technical Approach

The implementation separates page composition, visual styling, media, and UI feedback. Its working scope is the browser frontend; there is no application backend, database, or robot integration in this repository.

LAYER	STACK	PURPOSE
Frontend	React 18, TypeScript, Vite 5	Component rendering, source tooling, and frontend bundling
Routing	React Router DOM 6	Landing-page route and client-side not-found handling
Styling	Tailwind CSS 3, PostCSS, CSS variables	Responsive layouts, theme tokens, and visual treatments
Motion & media	lottie-react; IntersectionObserver; browser events	Hero media, viewport reveals, and scroll-driven cards
UI feedback	Sonner, Lucide; Radix-based UI primitives	Form notifications, icons, and shared UI infrastructure
State & assets	React state/effects; local public assets	Menu and form state, imagery, and animation loading

Design decisions that matter

Sections as composition boundaries. The route assembles focused components rather than a single monolithic page, keeping product content and interaction logic close together.

Browser-native presentation. Observers, scroll events, and local state power the experience without an animation service or server round trip.

Explicit implementation boundary. Inquiry feedback is a frontend demonstration. TanStack Query is provided in the shell, but no active data queries or mutations back these forms.

06 Value Delivered

From a robotics proposition to a coherent, explorable web presentation:

Branded	Immersive	Adaptable
PRODUCT IDENTITY	VISUAL STORYTELLING	FRONTEND STRUCTURE

- ✓ **A clear introduction to Atlas.** — The hero, product vision, and capability sections give visitors a common narrative instead of isolated technical claims.
- ✓ **A connected browsing surface.** — Product artwork, displayed details, and contact information sit within one responsive page, with navigation to the live sections.
- ✓ **Demonstrated inquiry interactions.** — Local validation, loading feedback, and confirmations make the intended form experience tangible without implying that leads are actually delivered.
- ✓ **A reusable presentation base.** — The component structure and themed styling provide an existing foundation for revising the product story and connecting future services.

Pulse Robot — a frontend product-marketing experience combining an Atlas brand narrative, animated visuals, responsive navigation, and browser-only inquiry demonstrations.